

1 Candidate relation layers



RNA co-expression



Methylation-expression



Copy-number co-perturbation

----- Zero graph $L_0 = 0$

2 Reliability gate

80% subsample
stability S_k

Cross-fitted score
 Δ_k vs L_0

$$u_k = S_k \max(\Delta_k, 0)$$

$\sum u_k > 0$
Eligible graphs
weighted

$\sum u_k = 0$
All graphs
rejected

A graph may be plausible yet ineligible.

3 Compact frozen signature

Graph-lasso ranking

Random-forest ranking

rank fusion

Freeze Top-20 genes,
directions and amplitudes

4 Outcome-free external test

Cohort-wise scaling
(no outcomes)

Apply frozen Top-20 score

External C-index + audit

**Auditable selection under
heterogeneous graph reliability**